Stochastic Calculus

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March 2026

$$dY_t = f_t\,dt + f_x\,dX_t + \frac{1}{2} f_{xx}\,dX_t^2$$

$$W_t \sim \mathcal{N}(0, t)$$

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Introduction

Modelling financial assets like stocks, bonds, commodities, etc. is a fundamental task in quantitative finance. When dealing with these assets, we often have to ask ourselves questions like:

- How can we incorporate the randomness of the market into our models?
- How can we model the evolution of asset prices over time?
- What mathematical tools can we use to create a realistic model?

The answer to these questions lies in the field of **stochastic calculus**, which provides a framework for modeling and analyzing systems that evolve over time with inherent randomness.

Introduction

Stochastic calculus is a branch of mathematics that extends traditional calculus to include random processes. Essentially it is a mix between:

- **Calculus**: The study of change and motion, which allows us to understand how systems evolve over time.
- **Probability Theory**: The study of randomness and uncertainty, which helps us model the unpredictable nature of real-world phenomena, such as financial markets.

Random Walk

We will start with the simplest model of a stochastic process: the **random walk**. Consider a random variable X_i that takes the following values:

$$X_i = \begin{cases} +\Delta x & \text{with probability } p = 0.5 \\ -\Delta x & \text{with probability } 1 - p = 0.5 \end{cases}$$

Usually, $\Delta x = 1$, but it can be any positive value. The random walk is defined as the cumulative sum of these random variables:

$$S_n = X_0 + \sum_{i=1}^{n} X_i$$

Where S_n represents the position of the random walk after n steps. X_0 is the initial position, which is often taken to be 0.

Random Walk

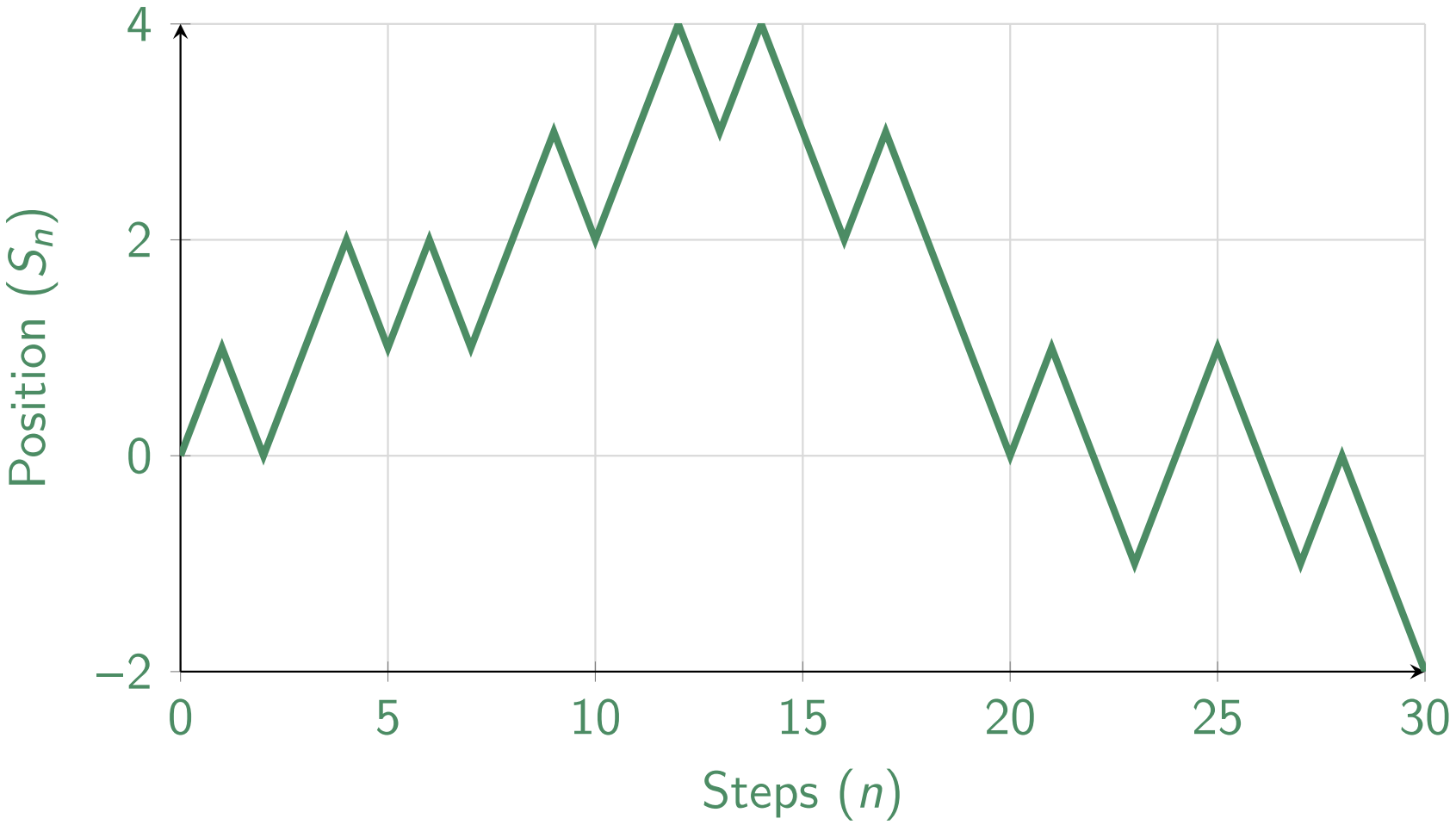

Figure: Random Walk path with $n = 30$, $X_0 = 0$, and $\Delta x = 1$.

Random Walk

The random walk is a simple yet powerful model that captures the essence of randomness in a process. It has several important properties:

- **Mean**: The expected value of S_n is zero, $E[S_n] = 0$.
- **Variance**: The variance of S_n grows linearly with the number of steps, $\mathrm{V}[S_n] = n \cdot (\Delta x)^2$.
- **Distribution**: As n increases, the distribution of S_n approaches a normal distribution due to the Central Limit Theorem.

Proof: Note that $E[X_i] = 0$ *and* $V[X_i] = (\Delta x)^2$*. Therefore,*
$E[S_n] = E[X_0] + \sum_{i=1}^{n} E[X_i] = 0$ *and* $V[S_n] = \sum_{i=1}^{n} V[X_i] = n \cdot (\Delta x)^2$*.*

Random Walk

The **Central Limit Theorem** states that as the number of steps n increases, the distribution of the random walk S_n becomes:

$$\frac{S_n - E[S_n]}{\sqrt{V[S_n]}} = \frac{S_n}{\sqrt{n \cdot (\Delta x)^2}} \xrightarrow{d} \mathcal{N}(0,1)$$

Therefore, when n is large, we obtain the following distribution for S_n:

Distribution of a Random Walk when $n \to \infty$

$$S_n \sim \mathcal{N}(0, n \cdot (\Delta x)^2)$$

Brownian Motion

While we could model a stock price as a random walk, it is often more convenient to use a **continuous-time model**, as financial assets move continuously rather than in discrete steps. By setting:

$$t = n \cdot \Delta t$$

we can now opperate in the continuous-time domain. By applying this change of variable (and making $n \to \infty$), we have that:

$$S_n \sim \mathcal{N}(0, \frac{t}{\Delta t} \cdot (\Delta x)^2)$$

Brownian Motion

The next step is to take the limit as $\Delta t \to 0$, making the process continuous in time. As we make Δt smaller, we need to adjust Δx accordingly to ensure that the variance remains finite.

Therefore we have to find a function f such that:

$$\Delta x = f(\Delta t)$$

and

$$\lim_{\Delta t \to 0} \frac{t}{\Delta t} \cdot (f(\Delta t))^2$$

remains finite.

Brownian Motion

Let's try some functions for $\Delta x = f(\Delta t)$:

- $\Delta x = 1$ then $\lim_{\Delta t \to 0} \frac{t}{\Delta t} \cdot (1)^2 = \lim_{\Delta t \to 0} \frac{t}{\Delta t} = \infty$
- $\Delta x = \Delta t$ then $\lim_{\Delta t \to 0} \frac{t}{\Delta t} \cdot (\Delta t)^2 = \lim_{\Delta t \to 0} t \cdot \Delta t = 0$
- $\Delta x = \Delta t^{1/3}$ then $\lim_{\Delta t \to 0} \frac{t}{\Delta t} \cdot (\Delta t^{1/3})^2 = \lim_{\Delta t \to 0} t \cdot \Delta t^{-1/3} = \infty$
- $\Delta x = \sqrt{\Delta t}$ then $\lim_{\Delta t \to 0} \frac{t}{\Delta t} \cdot (\sqrt{\Delta t})^2 = \lim_{\Delta t \to 0} t = t$

Therefore, the only function that allows us to have a finite variance in the limit is $\Delta x = \sqrt{\Delta t}$. This leads us to the definition of **Brownian Motion**.

Brownian Motion

$$W_t = \lim_{\Delta t \to 0} S_n \text{ with } \Delta x = \sqrt{\Delta t}$$

Brownian Motion

Brownian Motion (BM) lies in the heart of stochastic calculus and is a fundamental building block for modeling random processes in finance. Below we list some of the key properties of Brownian Motion:

Properties of Brownian Motion

- $W_0 = 0$ (starts at zero)
- W_t has independent increments
- W_t has normally distributed increments: $W_t - W_s \sim \mathcal{N}(0, t-s)$ for $0 \leq s < t$
- W_t has continuous paths, but is nowhere differentiable.

Brownian Motion

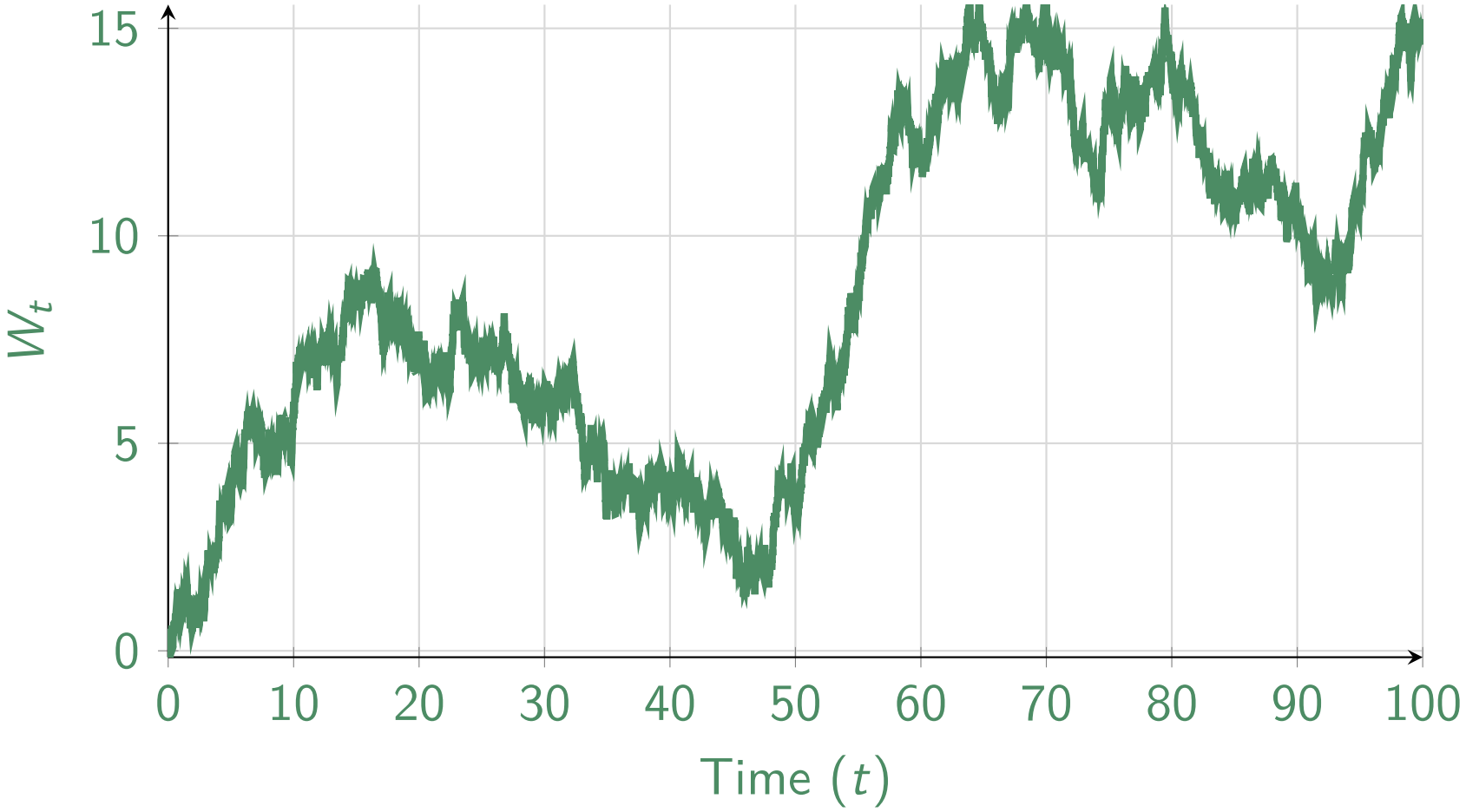

Figure: Sample path of Brownian Motion W_t.

Brownian Motion

One of the key properties of Brownian Motion is that is **nowhere differentiable**. As we will see later, this has important implications for how we can use calculus to analyze functions of BM.

An intuituive proof of this fact is the following: if we try to compute the derivative of W_t at any point, we would have to compute the limit:

$$\lim_{h \to 0} \frac{W_{t+h} - W_t}{h} = \lim_{h \to 0} \frac{\sqrt{h} \cdot Z}{h} = \lim_{h \to 0} \frac{Z}{\sqrt{h}} = \infty$$

where $Z \sim \mathcal{N}(0, 1)$. Therefore, the derivative does not exist at any point, and we say that W_t is nowhere differentiable.

Note that $W_{t+h} - W_t \sim \mathcal{N}(0, h)$, so we can write $W_{t+h} - W_t = \sqrt{h} \cdot Z$.

Brownian Motion

Another important property of BM is that **squared increments are non-zero**. In fact, we have that:

$$E[(dW_t)^2] = dt$$

or simply:

$$(dW_t)^2 = dt$$

Where dW_t represents the infinitesimal increment of BM at time t, and dt represents the infinitesimal increment of time. This differs from ordinary calculus, where we would have $(dx)^2 = 0$.

Proof: if we compute $E[(dW_t)^2] = V[dW_t] = dt$, *and* $V[(dW_t)^2] = E[(dW_t)^4] - (E[(dW_t)^2])^2 = 3dt^2 - dt^2 = 2dt^2$ *we see that as* $dt \to 0$, *the variance goes to zero faster than the mean, converging to* dt.

Functions of Brownian Motion

In finance, we can find that many quantities are often modeled as **functions of Brownian Motion**. In other words, we are interested in functions of the form:

$$X_t = f(t, W_t)$$

where X_t is a stochastic process, f is a deterministic function. Depending on the choice of f, we can model a wide variety of financial quantities such as:

- Stock prices
- Interest rates
- Volatility
- Option prices
- Credit spreads
- Exchange rates
- Bond prices
- Commodity prices
- Inflation rates

Functions of Brownian Motion

For example, if we want to model a stock price, we can use the following function of Brownian Motion:

$$X_t = X_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$$

Which is called **Geometric Brownian Motion** (GBM), where X_0 is the initial stock price, μ is the expected return, and σ is the volatility of the stock. We can also express GBM in terms of an stochastic differential equation (SDE):

$$dX_t = \mu X_t dt + \sigma X_t dW_t$$

Functions of Brownian Motion

It is more natural to express functions of BM in terms of SDEs, as they usually provide a more intuitive way to understand the dynamics of the process. For example, the SDE for GBM can be interpreted as follows:

$$\underbrace{\frac{dX_t}{X_t}}_{\text{Change in stock returns}} = \underbrace{\mu dt}_{\text{Deterministic expected return}} + \underbrace{\sigma dW_t}_{\text{Random shock}}$$

Despite gaining intuition from the SDE representation, we face the challenge of solving a differential equation that involves a stochastic term dW_t.

Functions of Brownian Motion

Some other examples of functions of Brownian Motion include:

- **Ornstein-Uhlenbeck**: $dX_t = \theta(\mu - X_t)dt + \sigma dW_t$, which models mean-reverting processes such as inflation or interest rates.
- **Cox-Ingersoll-Ross (CIR)**: $dX_t = \theta(\mu - X_t)dt + \sigma\sqrt{X_t}dW_t$, which models positive mean-reverting processes such as volatility.
- **Arithmetic BM (ABM)**: $dX_t = \mu dt + \sigma dW_t$, a linear function of BM used to model quantities that can take negative values.
- **Jump Diffusion (JD)**: $dX_t = \mu dt + \sigma dW_t + JdN_t$, where J represents the jump size and N_t is a Poisson process, used to model asset prices that can experience sudden jumps.

Functions of Brownian Motion

Figure: Sample paths of stochastic processes.

Itô's Lemma

Given a SDE as the ones we have seen before, how can we find the distribution of X_t? Let's say we have a SDE of the form:

$$dX_t = \mu X_t dt + \sigma X_t dW_t$$

If we try to compute the integral of both sides, we would have:

$$\int_0^t dX_s = \int_0^t \mu X_s ds + \int_0^t \sigma X_s dW_s$$

We clearly see that the right-hand side is not a simple Riemann integral, but a **stochastic integral**, which requires a different approach to solve.

Itô's Lemma

Instead of trying to solve the SDE directly via integration, we can use a powerful tool called **Itô's Lemma**, which essentially allows us to find the distribution of a function of BM without having to solve the SDE explicitly.

Itô's Lemma

Let X_t be a stochastic process. and $f(t,x)$ be a twice-differentiable function, then the process $Y_t = f(t, X_t)$ follows the SDE:

$$dY_t = f_t dt + f_x dX_t + \frac{1}{2} f_{xx} (dX_t)^2$$

where f_t is the partial derivative of f with respect to t, f_x is the partial derivative of f with respect to x, and f_{xx} is the second partial derivative of f with respect to x.

Itô's Lemma

At first glance, Itô's Lemma may look unintuitive, and it is not immediately clear how or why it works. Let's try to understand the intuition behind it. Consider the following function:

$$y = f(x)$$

If we want to find the change in y when x changes by a small amount dx (also known as the differential, dy), we can use the Taylor expansion:

$$dy = f_x dx + \frac{1}{2} f_{xx} (dx)^2 + \dots$$

Itô's Lemma

In ordinary calculus, we would have that $(dx)^2 = 0$, so we would simply have

$$dy = f_x dx$$

Consider now the case when $y = f(t, x)$, then we would have:

$$dy = f_t dt + f_x dx$$

Now let $x = X_t$ be a stochastic process and $y = Y_t = f(t, X_t)$, if we try to apply the same logic, we would have:

$$dY_t = f_x dX_t + f_t dt + \frac{1}{2} f_{xx} (dX_t)^2 + \frac{1}{2} f_{tt} (dt)^2 + \dots$$

Itô's Lemma

It is clear that the terms involving $(dt)^2$ and higher powers of dt will go to zero as $dt \to 0$. However, the term involving $(dX_t)^2$ does not go to zero. Recall that we have seen that

$$(dW_t)^2 = dt$$

Therefore, this implies that if $X_t = f(W_t, t)$, then

$$(dX_t)^2 \neq 0$$

This leads to the following expression which is indeed Itô's Lemma:

$$dY_t = f_t dt + f_x dX_t + \frac{1}{2} f_{xx} (dX_t)^2$$

Itô's Lemma

Now that the intuition behind Itô's Lemma is clear, let's see a practical example of how to use it. There are two main usecases for Itô's Lemma:

1. Given a function $Y_t = f(t, X_t)$, find the SDE that Y_t follows.
2. Given an SDE for X_t, find the distribution of $Y_t = f(t, X_t)$.

We will start with number 1. Suppose we have a function

$$f(x, t) = x_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma x}$$

which is the solution to the GBM SDE. How can we get the SDE back? In other words, given $X_t = f(t, W_t)$, how can we find the SDE that X_t follows (the expression for dX_t)?

Itô's Lemma

We know that $X_t = f(t, W_t)$, therefore:

$$X_t = X_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$$

The partial derivatives of f, evaluated at (t, W_t), are:

$$\begin{cases} f_t & = X_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t} \cdot (\mu - \frac{1}{2}\sigma^2) = X_t \cdot (\mu - \frac{1}{2}\sigma^2) \\ f_x & = X_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t} \cdot \sigma = X_t \cdot \sigma \\ f_{xx} & = X_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t} \cdot \sigma^2 = X_t \cdot \sigma^2 \end{cases}$$

Itô's Lemma

Now, if we apply Itô's Lemma

$$dX_t = f_t dt + f_x dW_t + \frac{1}{2} f_{xx} (dW_t)^2$$

We have that:

$$dX_t = X_t \cdot (\mu - \frac{1}{2}\sigma^2) dt + X_t \cdot \sigma dW_t + \frac{1}{2} X_t \cdot \sigma^2 dt$$

Which simplifies to the original SDE for GBM:

$$dX_t = \mu X_t dt + \sigma X_t dW_t$$

Itô's Lemma

Now we ask ourselves the second question: given an expression for dX_t, how can we find the distribution of X_t? Let's say we have the previous SDE:

$$dX_t = \mu X_t dt + \sigma X_t dW_t$$

Itô's Lemma allows us to find the distribution of $Y_t = f(t, X_t)$. Once we have it, we can easily find the distribution of X_t by applying the inverse function of f to Y_t as follows:

$$X_t = f^{-1}(t, Y_t)$$

Therefore, choosing the right function f is crucial to be able to find the distribution of X_t.

Itô's Lemma

Let's start by computing dY_t, using Itô's Lemma:

$$dY_t = f_t dt + f_x dX_t + \frac{1}{2} f_{xx} (dX_t)^2$$

We know that $dX_t = \mu X_t dt + \sigma X_t dW_t$, therefore:

$$dY_t = f_t dt + f_x(\mu X_t dt + \sigma X_t dW_t) + \frac{1}{2} f_{xx} (\mu X_t dt + \sigma X_t dW_t)^2$$

Recall that $(dW_t)^2 = dt$, therefore:

$$dY_t = f_t dt + f_x(\mu X_t dt + \sigma X_t dW_t) + \frac{1}{2} f_{xx} (\sigma X_t)^2 dt$$

Rearranging terms, we have:

$$dY_t = (f_t + \mu X_t f_x + \frac{1}{2} \sigma^2 X_t^2 f_{xx}) dt + \sigma X_t f_x dW_t$$

Itô's Lemma

If we integrate both sides, from 0 to t, we have:

$$Y_t = Y_0 + \int_0^t (f_t + \mu X_s f_x + \frac{1}{2}\sigma^2 X_s^2 f_{xx}) ds + \int_0^t \sigma X_s f_x dW_s$$

What follows is to choose the function f such that the partial derivatives evaluated in (s, X_s) simplify the integrals. We want to choose f so that:

1. f^{-1} exists. Therefore f has to be a bijection.
2. f_x is be proportional to $\frac{1}{x}$ so when we evaluate it at (s, X_s), we get a constant that can be taken out of the integral.
3. f_{xx} is be proportional to $\frac{1}{x^2}$ so when we evaluate it at (s, X_s), we get a constant that can be taken out of the integral.
4. f_t ideally is zero, so that the first term of the integral disappears

Itô's Lemma

The function that satisfies all these conditions is:

$$f(t,x) = \ln(x)$$

The partial derivatives of f are:

$$\begin{cases} f_t & = 0 \\ f_x & = \frac{1}{x} \\ f_{xx} & = -\frac{1}{x^2} \end{cases}$$

Now, if we evaluate the partial derivatives at (s, X_s), we have:

$$\begin{cases} f_t(s,X_s) & = 0 \\ f_x(s,X_s) & = \frac{1}{X_s} \\ f_{xx}(s,X_s) & = -\frac{1}{X_s^2} \end{cases}$$

Itô's Lemma

Now we can substitute these values into the expression for Y_t:

$$Y_t = Y_0 + \int_0^t (0 + \mu X_s \cdot \frac{1}{X_s} + \frac{1}{2}\sigma^2 X_s^2 \cdot -\frac{1}{X_s^2})ds + \int_0^t \sigma X_s \cdot \frac{1}{X_s} dW_s$$

Simplifying, we have:

$$Y_t = Y_0 + \int_0^t (\mu - \frac{1}{2}\sigma^2)ds + \int_0^t \sigma dW_s$$

Which is easy to solve, and the final expression for Y_t is:

$$Y_t = Y_0 + (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$$

Itô's Lemma

Now that we have the distribution of Y_t, we can find the distribution of X_t by applying the inverse function of f to Y_t:

$$X_t = f^{-1}(t, Y_t) = e^{Y_t} = e^{Y_0 + (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$$

And defining $X_0 = e^{Y_0}$, we have the final expression for X_t:

$$X_t = X_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$$

Therefore, solving SDEs via Itô's Lemma consists of finding the right function f that allows us to simplify the integrals so we can find the distribution of Y_t, and then apply the inverse function to find the distribution of X_t.

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